CE205

Lecture 3 Activities

Activity 1: Consider a relation **R {title, year, studioName, president, presAddr}**

which is abbreviated as **R (T, Y, S, P, A).**

That is, each tuple in this relation describes a movie, its studio, the studio president, and the president's address.

Convert R to BCNF?

Guideline:

i. Identify FDs

ii. Using closure of attributes, find candidate key (s)

iii. Check whether FDs are violating BCNF

iv. Decompose those FDs into new relations

Activity 2: Use this relation to answer the questions during lectures

STUDENT_ACTIVITY

StudentID	StudentName	Activity	ActivityFee	AmountPaid
100	Jones	Golf	65.00	65.00
100	Jones	Skiing	200.00	0.00
200	Davis	Skiing	200.00	0.00
200	Davis	Swimming	50.00	50.00
300	Garrett	Skiing	200.00	100.00
300	Garrett	Swimming	50.00	50.00
400	Jones	Golf	65.00	65.00
400	Jones	Swimming	50.00	50.00

Activity 3: Consider a relation **R(A, B, C, D)** with the following set of FDs.

AB → C

C→ D

D → A

Why is the relation R not in BCNF? Decompose the relation, as necessary, into a collection of sub-relations in BCNF.

Guideline:

Activity 4: Consider relation **R (A, B, C, D)** with MVDs A→→B and A→→C

i) Find all the 4NF violations.
ii) Decompose the relation into a collection of sub-relations in 4NF.